INVISIBILITY!

ENCRYPT, OBFUSCATE, ATTACK ...

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It's not that most black hat hackers have a love for the theatrics that they don the black mask and the blazer, obscuring their identity is paramount towards being able to continue with their activities. The shroud of invisibility often lets them carry out their nefarious (or vigilante) acts with impunity. And it can be easily argued that the whistleblowers of the current digital era are enabled thanks to the levels of privacy afforded by the very same technologies. While we aren't taking any sides, we dug around a little to figure out how they go about their activities, how they hide their traces, how they stay off the grid, how they dish out their blend of activism. In no manner should this list be considered complete. Here's how they do it.

SEPARATE IDENTITIES

Some things are a no-brainer. Almost every black hat that was caught and the countless ones that are yet to be caught operate under a pseudonym. And it's not just the name, each identity is only utilised with a certain set of hardware. On your work laptop, he/she might be a John or a Jane and on another machine they might be a genderless X4yn3r. Consistency is important, those who're out to track you down and discern your identity are always getting some bit of information or the other which they're piecing together which is also why black hats tend not to use the same software for long, because no matter how random a software can claim to be, there's always a pattern and over time, the more they see the easier it becomes to narrow down on someone.

It's when the identities mix, that they tend to get caught. Ross Ulbricht a.k.a. Dread Pirate Roberts was caught when he gave out his personal email ID while astroturfing. The FBI had a lot of data about his identities and this one post on a forum allowed them to connect the dots and nail him. The importance of maintaining separate identities is a matter of life or death.

RATTINGS

Remote Access Trojans are how black hats gain control over your machine. And that's also how they add computers to their personal botnets. Which incidentally become staging areas for attacks and sending unsolicited emails. Opening a simple email is all it takes for the payload to be delivered and for a backdoor to be opened. From that point onwards, you'll have no idea as to what nefarious purpose you machine was used for. Say hi to the feds for us.

VPN

The go to tool when it comes to hiding your internet traffic from snoopers, the all popular VPN is heavily advised but little is known about the efficacy of VPNs. Sure they allow you to watch geo-restricted content on YouTube but they aren't great for ensuring privacy in their default usage scenarios. While many VPNs claim to not keep logs, they're also signatories to laws that require them to lie about keeping logs once contacted by a security agency to do so. Little white lies...

Hackers do their homework and check if a VPN is actually obfuscating traffic by using tools like WireShark to check if switching on a VPN breaks existing connections. Then they check if the VPN is indeed re-routing all the traffic from the machine or just a portion of it limited to browser sessions and specific apps. This matters because existing browser sessions and applications on your computer will immediately try to ping back home when a network connection is modified and if any of these pingbacks are not routed through the VPN, then it's as good as not using a VPN.

A general trend observed among hackers who speak up is to use multiple VPN services and switching between them. And when it comes to paying for these services, the methods of payment are kept anonymous. When a VPN service becomes sufficiently popular, the feds bring in listening equipments and monitor the end points, this is one manner in which traffic correlation is done in order and it renders a VPN useless. And it's not just the endpoints, even network routers along the way are tapped to perform man in the middle attacks and generate logs. Most governmental security programs focus on always gathering metadata until some suspicious activity is flagged and then the metadata is mined to generate a profile. Hence, the need for using multiple VPN services. And lastly,